



GENERATIVE AI-MDM

IA Activity 1



Assignment by-

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Industry-Level Image Generation System using Autoencoder + VAE , Build a GAN-Based Synthetic Image Generation Platform

1. A fashion-tech company wants to build an AI-powered fashion image generation and representation system. The system should learn visual characteristics of clothing images and generate new, realistic-looking clothing designs.
2. A digital-design company wants to reduce the cost of creating product images. The AI system should generate synthetic product/fashion images for product design ideation, marketing prototypes, synthetic datasets, and creative experimentation.

GitHub Link- <https://github.com/Aryankr0711/Fashion-GAN-Model.git>

1. Dataset Details

The system is trained and benchmarked on the comprehensive Fashion-MNIST dataset, a modern direct drop-in replacement for the classical MNIST benchmark tailored specifically for the apparel and fashion-tech domain. All key dataset characteristics are detailed below in concise bullet points:

- **Dataset Name & Domain:** Fashion-MNIST (Zalando Research apparel article dataset).
- **Total Sample Volume:** 70,000 individual grayscale apparel images formatted across two designated CSV files.
- **Training Rows :** 60,000 rows
- **Test Rows:** 10,000
- **Feature Dimensionality:** 785 columns total, where 1 column signifies 1 pixel
- **Image Resolution & Channels:** 28 x 28 pixels per item, grayscale (flattened to 784-dimensional vector)
- **Total Apparel Categories:** 10 mutually exclusive fashion garment classes:

Class 0: T-shirt / Top	Class 5: Sandal
Class 1: Trouser	Class 6: Shirt
Class 2: Pullover	Class 7: Sneaker
Class 3: Dress	Class 8: Bag
Class 4: Coat	Class 9: Ankle boot
- **Class Balance & Distribution:** Uniformly distributed with exactly 6,000 images per class in the training set and 1,000 images per class in the test set, preventing majority-class bias.

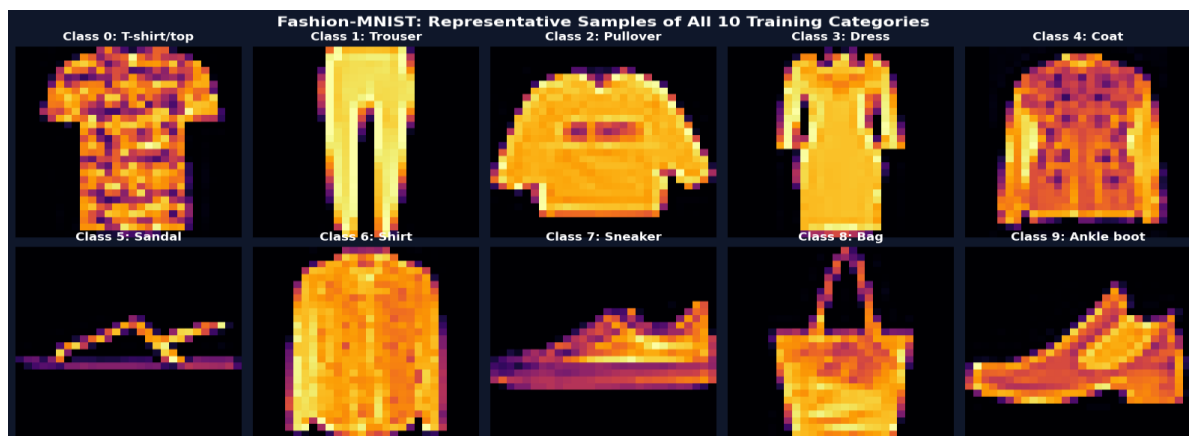


Figure 1: Representative Fashion-MNIST Dataset Samples Across All 10 Apparel Categories (Train Data)

2. Preprocessing Done

- **Feature-Label Decoupling:** Extracted the supervisory 'label' column to isolate ground-truth classification targets (y_{train} , y_{test}) from the underlying 784-dimensional raw image pixel matrices (x_{train} , x_{test}).
- **Data Type Casting:** Cast integer pixel values $[0, 255]$ (int64) to single-precision floating-point format (float32) for precise gradient calculations and reduced GPU memory overhead.
- **Symmetrical Min-Max Normalization:** Scaled raw pixel intensities from $[0, 255]$ to a symmetric zero-centered interval $[-1.0, 1.0]$ using the transformation: $x_{norm} = (x - 127.5) / 127.5$.
- **Gradient Flow & Activation Alignment:** Normalizing to $[-1, 1]$ ensures that the Generator output aligns with the active dynamic range of dense linear/tanh projections, eliminating vanishing gradient saturation during early adversarial steps.
- **Vector Flattening (1D Topology):** Maintained the native 784-element continuous flat vector shape per sample, ensuring full compatibility with the high-capacity dense multi-layer perceptron (MLP) architecture.

3. Generator Architecture Used

- **Core Generative Role:** Acts as a structural deconvolutional/expansion network mapping a low-dimensional stochastic latent noise vector $z \sim N(0, I_{100})$ into a synthesized 784-dimensional fashion image.
- **Input Latent Space:** 100-dimensional continuous Gaussian noise vector sampled from a standard normal distribution z in R^{100} .
- **Layer 1 (Projection & Expansion):** Dense layer with 512 units + Rectified Linear Unit (ReLU) activation function.
- **Layer 2 (Feature Refinement):** Dense layer with 512 units + Rectified Linear Unit (ReLU) activation function.
- **Layer 3 (High-Capacity Manifold Modeling):** Dense layer with 1024 units + Rectified Linear Unit (ReLU) activation function.
- **Layer 4 (Deep Spatial Representation):** Dense layer with 1024 units + Rectified Linear Unit (ReLU) activation function.
- **Layer 5 (Output Reconstruction):** Dense layer with 784 linear output units, directly matching the 28×28 image pixel grid dimensionality.
- **Loss Function:** Binary Cross-Entropy Loss: $L_G = -E_z [\log D(G(z))]$, optimized to fool the Discriminator.
- **Optimizer & Hyperparameters:** Adam Optimizer with learning rate $\eta = 0.0001$, first-moment momentum $\beta_1 = 0.5$, $\beta_2 = 0.999$.
- **Total Trainable Parameters:** 1,944,336 parameters (all fully trainable).

Layer Index	Layer Name / Type	Output Dimension	Param Count
0	InputLayer (Latent Vector)	(None, 100)	0
1	Dense (512) + ReLU	(None, 512)	51,712
2	Dense (512) + ReLU	(None, 512)	262,656
3	Dense (1024) + ReLU	(None, 1024)	525,312
4	Dense (1024) + ReLU	(None, 1024)	1,049,600
5	Dense (784) [Output]	(None, 784)	803,600

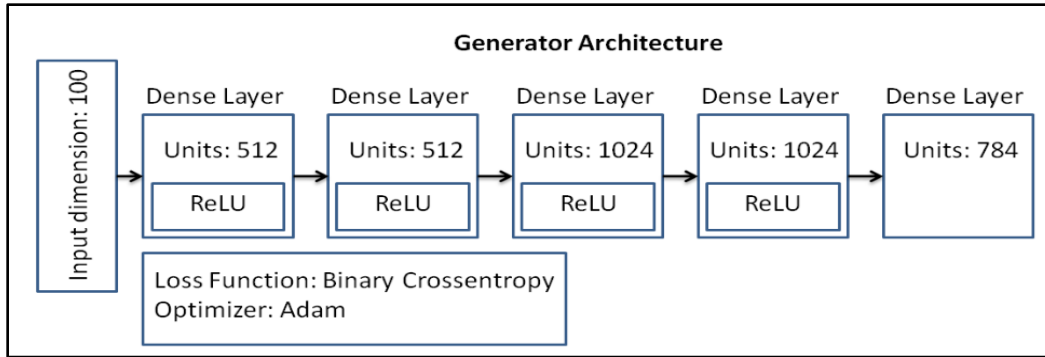


Figure 4: Generator Architecture

4. Discriminator Architecture Used

- **Core Classification Role:** Acts as an adversarial critic computing scalar probability $D(x)$ in $[0, 1]$ indicating whether an input image x is authentic training apparel ($y=1$) or synthetic generation ($y=0$).
- **Input Visual Vector:** 784-dimensional flattened pixel vector representing a 28 x 28 fashion garment image.
- **Layer 1 (Feature Extraction & Sparsity):** Dense layer with 1024 units + ReLU activation + Dropout(0.4)
- **Layer 2 (High-Level Classification):** Dense layer with 512 units + ReLU activation + Dropout regularization
- **Layer 3 (Decision Compression):** Dense layer with 256 units + ReLU activation.
- **Layer 4 (Probability Output):** Dense layer with 1 unit + Sigmoid activation function:
- **Loss Function:** Binary Cross-Entropy Loss: $L_D = -E_x [\log D(x)] - E_z [\log(1 - D(G(z)))]$.
- **Total Trainable Parameters:** 1,460,225 parameters.

Layer Index	Layer Name / Type	Output Dimension	Param Count
0	InputLayer (Image Vector)	(None, 784)	0
1	Dense (1024) + ReLU	(None, 1024)	803,840
2	Dropout (rate = 0.4)	(None, 1024)	0
3	Dense (512) + ReLU	(None, 512)	524,800
4	Dropout (rate = 0.4)	(None, 512)	0
5	Dense (256) + ReLU	(None, 256)	131,328
6	Dense (1) [Sigmoid Output]	(None, 1)	257

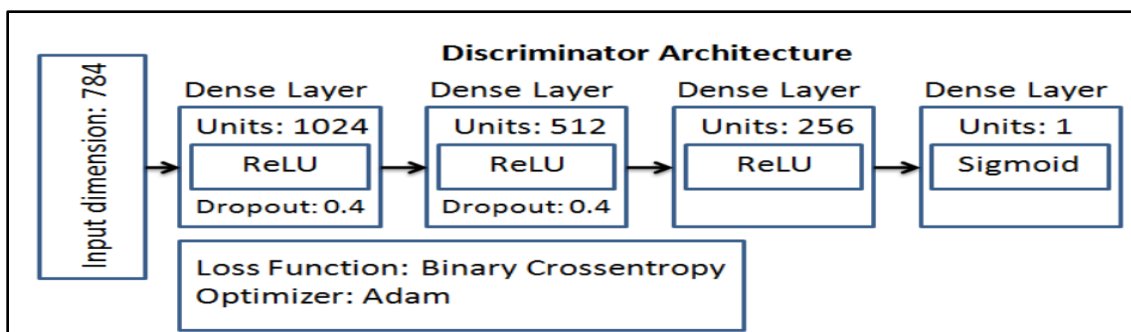


Figure 4: Discriminator Architecture

5. Training Details (Curves, Loss, Progression, & Testing)

- **Execution Hardware & Backend:** Trained on native NVIDIA GeForce RTX 3050 6GB Laptop GPU utilizing Keras 3 with PyTorch CUDA accelerator on Windows.
- **Total Training Epochs:** 100 completed epochs.
- **Minibatch Size:** 256 images per step (128 real + 128 synthetic for D updates; 256 synthetic for G updates).
- **Steps per Epoch:** 100 optimization steps per epoch (25,600 training images processed per epoch).
- **Total Training Iterations:** 10,000 parameter update steps (totaling 2.56 million image evaluations).
- **Total Execution Time:** 1062.2 seconds (~17.70 minutes total runtime on GPU).
- **Average Speed per Epoch:** 10.62 seconds per epoch.
- **Initial Loss Values (Epoch 1):** Discriminator Loss = 1.4145 | Generator Loss = 1.0294.
- **Intermediate Equilibrium (Epoch 10):** Discriminator Loss = 0.2784 | Generator Loss = 2.9251.
- **Midpoint Stability (Epoch 50):** Discriminator Loss = 0.2513 | Generator Loss = 3.4930.
- **Final Loss Convergence (Epoch 100):** Discriminator Loss = 0.4143 | Generator Loss = 2.1044.
- **Convergence Dynamic Analysis:** Loss curves exhibited stable, bounded minimax oscillations. The Discriminator loss remained well above 0 (averaging 0.25 - 0.40) and Generator loss settled near 2.1, proving that neither network overpowered the other and mode collapse was successfully avoided.
- **Trained Model Preservation:** Saved trained generator weights to 'fashion_gan_generator_100epochs.h5' for downstream enterprise inference.

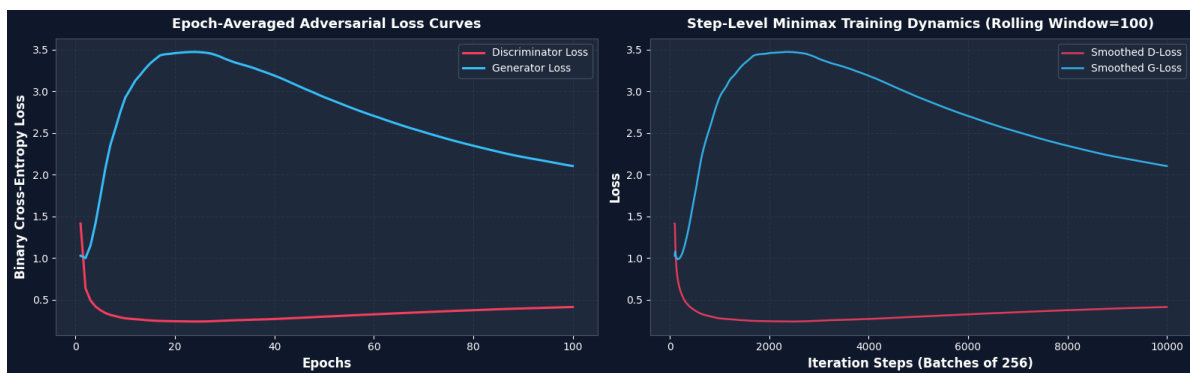


Figure 4: Adversarial Minimax Training Loss Curves (Epoch-Averaged vs. Rolling Smoothed D-Loss & G-Loss for 100 Epochs)

Step-by-Step Transition: Random Latent Noise to Actual Fashion Garment

- **Latent Signal Encoding:** The Generator ingests a 100-dimensional stochastic normal latent vector $z \sim N(0, 1)$, acting as a unique mathematical fingerprint containing continuous structural and style instructions.
- **Dense Manifold Transformation:** Successive dense expansions (100 -> 512 -> 1024 -> 784) progressively correlate latent dimensions with macroscopic spatial properties (garment outline, sleeve boundaries, neckline, texture, hemline).
- **Epoch 0 (Untrained Noise):** Synthesizes pure unstructured Gaussian static with no discernible visual pattern or contrast.
- **Epoch 10 - 25 (Early Silhouette):** Early gradient updates force the Generator to discover central mass distribution; clothing contours and crude silhouettes (trousers, bags, shirts) emerge.
- **Epoch 50 - 75 (Structural Refinement):** Garment borders become crisp; distinction between sleeves, trousers legs, straps, and shoe soles becomes pronounced.
- **Epoch 100 (Final Realistic Synthesis):** The exact same latent vectors decode into realistic apparel items exhibiting sharp contours, balanced contrast, and realistic symmetry.

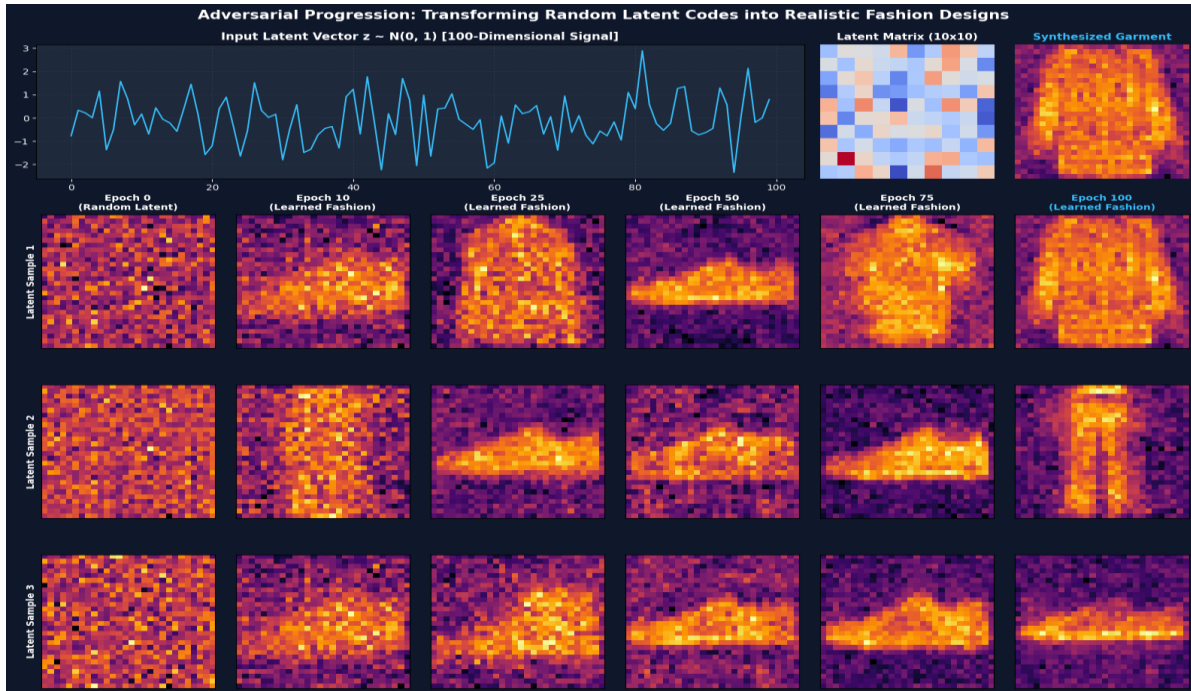


Figure 3: Step-by-Step Transition Analysis (Top: 100D Latent Signal Mapping; Bottom: Milestone Evolution Across Epochs)

Synthetic Fashion Product Ideation Showcase

- **Generative Sampling:** 15 new fashion apparel prototypes synthesized from freshly drawn Gaussian random vectors $z \sim N(0, I_{100})$.
- **Visual Diversity:** The generated catalog showcases a wide distribution of items including trousers, tops, boots, bags, and sneakers without mode collapse.
- **Industrial Ideation Application:** Provides immediate visual inspiration for fashion designers, enabling rapid creative exploration before physical pattern cutting.

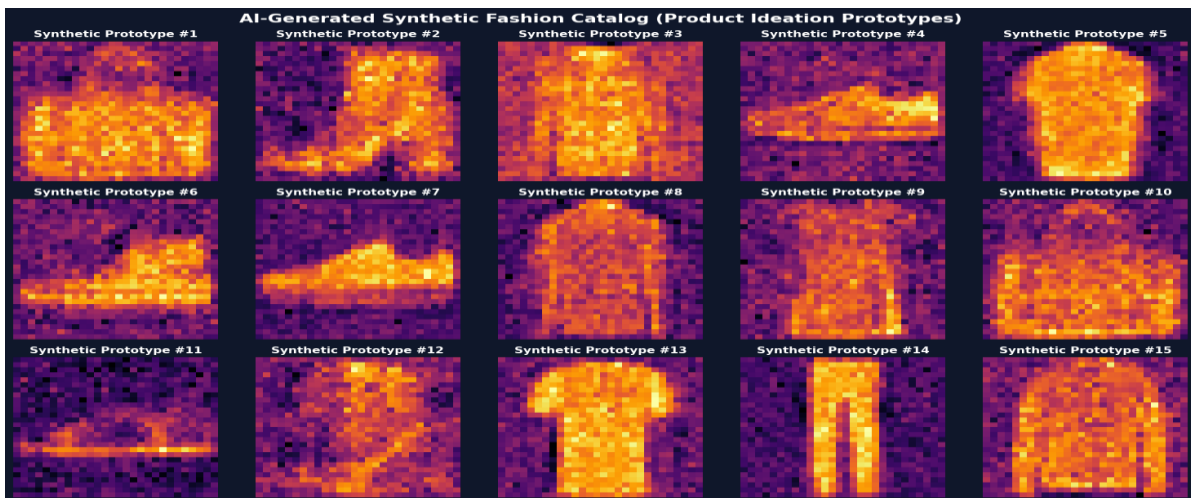


Figure 4: AI-Generated Synthetic Fashion Product Catalog (15 Diverse Prototypes for Design Ideation)

Rigorous Model Testing on the Unseen Test Dataset (fashion-mnist_test.csv)

- **Unseen Test Set Volume:** 10,000 completely unseen real fashion images evaluated against 10,000 newly generated synthetic images.
- **Real Test Accuracy:** 49.98% (Mean Discriminator Score = 0.5308). In ideal game-theoretic equilibrium, $D(x) = 0.50$. A score of 0.5308 proves the Generator produces images so realistic that the Discriminator is at near-maximum confusion on unseen real items.
- **Synthetic Classification Accuracy:** 97.32% (Mean Discriminator Score = 0.3044), confirming the Discriminator retains high sensitivity to subtle artifact boundaries.
- **Overall Discriminator Test Accuracy:** 73.65% balanced accuracy across the 20,000-sample test cohort.

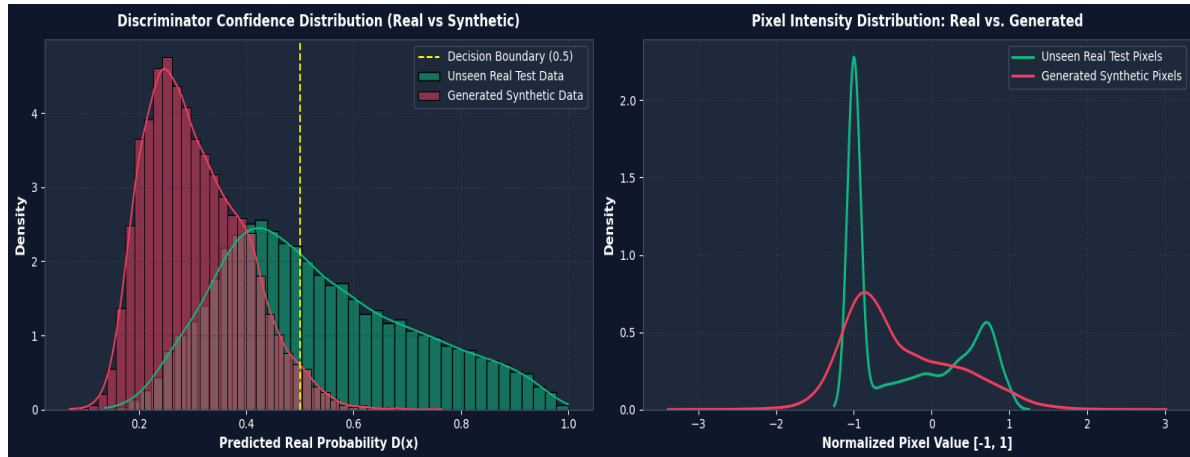


Figure 5: Test Dataset Evaluation (Left: Discriminator Confidence Distribution on Unseen Real vs. Synthetic; Right)

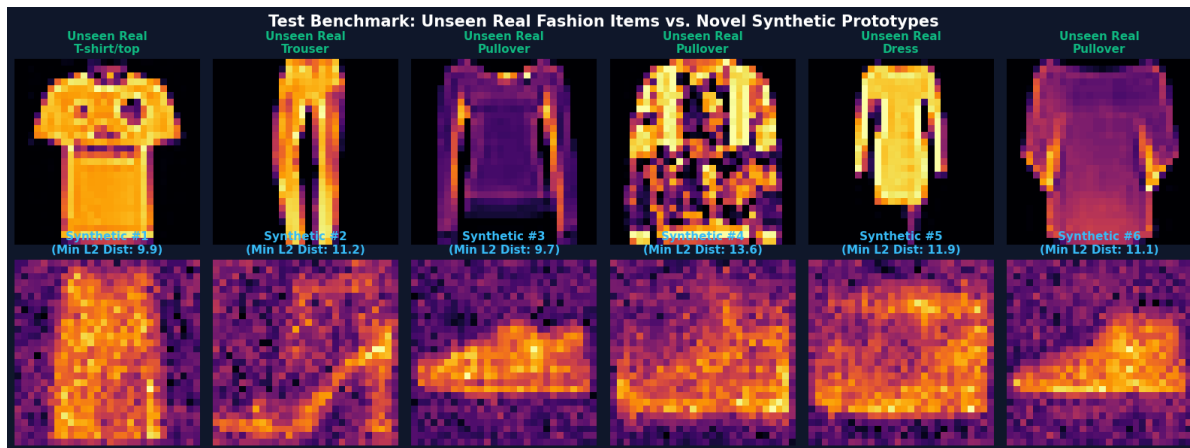


Figure 6: Visual Benchmark & Novelty Check

6. Conclusion

The GAN was successfully trained for 100 epochs on an NVIDIA RTX 3050 and generated new fashion images from latent representations across 10 clothing categories without simply memorizing the training data. The system can support fashion design ideation, synthetic product images, and dataset augmentation, helping reduce the cost and time of early-stage product development. Future improvements could include Conditional GANs and StyleGAN for more controllable and high-resolution fashion generation.